MPPT OVERALL WORKING PRINCIPAL

12V/24V MODE

AUTO DETECT 12V CHARGING MODE OR 24V CHARGING MODE ON EVERY STARTUP.

FAN CONTROLLING AND OVER HEAT PROTECTION

AUTOMATIC TEMPERATURE CONTROLLING THROUGH COOLING FAN.

WHEN TEMPERATURE EXCEED SAFETY LEVEL CHARGING STOPS, WARNING LED(RED) STARTS BLINKING AND A MESSAGE SHOWING **"WARNING!!! OVER HEAT"** APPEARS ON LCD SCREEN. THE CHARGING REMAINS STOP TILL THE TEMPERATURE DECREASES BELOW SAFETY LEVEL.

HIGH VOLTAGE PROTECTION

WHEN SOLAR VOLTAGE EXCEEDS 150V(MODEL #: EXO-M-100)
CHARGING STOPS, WARNING LED(RED) STARTS BLINKING AND A MESSAGE SHOWING
"WARNING!!! HIGH PV VOLATGE" APPEARS
REMAINS STOP TILL THE VOLTAGE DECREASES BELOW SAFETY LEVEL.

OVER CHARGE PROTECTION

WHEN BATTERY VOLTAGE EXCEEDS 15V (12V CHARGING MODE) OR 30V (24V CHARGING MODE) CHARGING STOPS, WARNING LED(RED) STARTS BLINKING AND A
MESSAGE SHOWING **"WARNING!!! OVER CHARGING"** APPEARS ON LCD SCREEN. THE CHARGING REMAINS STOP TILL THE VOLTAGE DECREASES BELOW SAFETY LEVEL.

BATTERY LOW WARNING

WHEN BATTERY CAPACITY DECREASES BELOW 20%, WARNING LED(RED) STARTS BLINKING AND A MESSAGE SHOWING **"WARNING!!! LOW BATTERY"** APPEARS ON LCD SCREEN. THE MESSAGE CONTINUES TO APPEAR ON LCD TILL THE CAPACITY RISES ABOVE 20%.

WHEN BATTERY CAPACITY DECREASES BELOW 50%, MPPT SWITCHES FROM **"SOLAR CHARGING MODE"** TO **"HYBRID CHARGING MODE".**

WHEN BATTERY REACHES 100% CAPACITY, MPPT RETURNS TO **"SOLAR CHARGING MODE".**

STAND BY MODE

WHEN SOLAR ENERGY IS INSUFFICIENT TO CHARGE THE BATTERY, MPPT SWITCHES TO **"STAND BY MODE"**, HYBRID MODE LED(BLUE) TURNS ON AND A MESSAGE SHOWING **"PV INTERRUPTED"** APPEARS ON LCD SCREEN.

IN **"STAND BY MODE"** GRID SUPPLY IS TURNED ON SO THAT BATTERY CAN BE CHARGED THROUGH GRID.

SOLAR CHARGING MODE

WHEN SOLAR ENERGY IS SUFFICIENT TO CHARGE THE BATTERY, MPPT REMAINS IN **"SOLAR CHARGING MODE".**

WHEN MPPT IS CHARGING THE BATTERY THROUGH SOLAR POWER, SOLAR MODE LED(Y REMAINS ON.

3. BACK TO SOLAR MODE VOLTAGE

   SELECTABLE VOLTAGE WHEN TO SHIFT FROM HYBRID/UTILITY MODE TO SOLAR MODE i.e., TURN OFF UPS/INVERTER SUPPLY

4. BACK TO HYBRID/UTILITY MODE VOLTAGE

   SELECTABLE VOLTAGE WHEN TO SHIFT FROM SOLAR MODE TO HYBRID/UTILITY MODE i.e., TURN ON UPS/INVERTER SUPPLY

5. DC LOAD OFF VOLTAGE

   SELECTABLE VOLTAGE TO TURN OFF/DISCONNECT DC LOAD FROM BATTERY

6. DC LOAD ON VOLTAGE

   SELECTABLE VOLTAGE TO TURN ON/RECONNECT DC LOAD FROM BATTERY

7. LOW BATTERY WARNING & ALARM VOLTAGE

   SELECTABLE LOW BATTERY WARNING & ALARM VOLTAGE

8. LOW BATTERY ALARM ON/OFF

   SELECTABLE OPTION TO TURN LOW BATTERY ALARM ON OR OFF

8. ERROR MESSAGE

WHEN A MALFUNCTION OCCURS, AN ERROR MESSAGE WILL BE DISPLAYED ON THE LCD SCREEN. FAULTS INCLUDE SYSTEM FAULTS AND MPPT FAULTS.

IN SOME CASES, YOU MAY BE ADVISED TO CONTACT **EXOTECH**, PLEASE PROVIDE THE FOLLOWING INFORMATION.

INFORMATION ABOUT THE MPPT:

- MODEL NUMBER
- ERROR MESSAGE ON THE LCD
- A SHORT DESCRIPTION OF THE PROBLEM
- DC INPUT VOLTAGE
- CAN YOU REPRODUCE THE FAILURE? IF SO, HOW?
- HAS THIS PROBLEM OCCURRED IN THE PAST?
- WHAT WERE THE ENVIRONMENTAL CONDITIONS WHEN THE PROBLEM OCCURRED?

INFORMATION ABOUT THE PHOTOVOLTAIC PANELS:

- PV PANEL MANUFACTURER'S NAME AND MODEL
- PANEL OUTPUT POWER
- VOC OF THE PANEL
- VMP OF THE PANEL
- IMP OF THE STRING/ARRAY
- THE NUMBER OF PANELS IN EACH STRING